Econ 814 Recitation 7 Notes

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1 University of Oslo Fall 2014 Exercise B

Consider a model in which there are two periods ($t = 1, 2$) and a unit mass of identical agents. In period 2 there are two states, denoted by s_G and s_B. The state turns out to be s_G with probability $p \in (0, 1)$ and thus state s_B happens with probability $1 - p$. Each agent receives income e_1 in period 1 and $e_2(s)$ in state $s \in \{s_G, s_B\}$ of period 2, where $e_2(s_G) \geq e_2(s_B)$. All households (you can think of them as a single representative household) have the same preferences over consumption

$$U = \frac{c_1^{1-\gamma}}{1-\gamma} + \beta E[\frac{c_2(s)^{1-\gamma}}{1-\gamma}]$$

where $\gamma \geq 0$, $0 < \beta \leq 1$, and E denotes the expectation operator with respect to the state s. All markets are competitive. Households can buy a bond, b, at price 1 in period 1 which pays an (endogenous) interest $1 + r$ in period 2 and is in zero supply. Households start with initial assets of zero, that is they have no bond holdings initially. Note that there is no capital in this economy. There are also no firms (income is obtained by fishing).

1.1 Part a

(5 Points) Write down the households state-by-state budget constraints for both periods (you can assume the constraints hold with equality). Solution: The budget constraints read

$$c_1 + b = e_1$$

$$c_2(s) = e_2(s) + (1 + r)b, \forall s \in \{s_G, s_B\}$$

1.2 Part b

(5 Points) Show that the households' constrained optimization problem is equivalent to maximizing the objective function

$$U = \frac{(e_1 - b)^{1-\gamma}}{1-\gamma} + \beta E[\frac{(e_2(s) + (1 + r)b)^{1-\gamma}}{1-\gamma}]$$

with respect to the bond holdings, b. (Hint: no proof is required here, just state the procedure of how to derive the above objective function) Solution: The households' optimization problem is to maximize U in Equation (6) subject to the period-by-period budget constraints. The objective function is derived by simply reducing consumption in U with the period-by-period budget constraints.

1.3 Part c

(10 Points) Derive the optimality condition for the households' bond holdings and state as well the the bond market clearing condition. What are the implications of the bond market clearing condition for the equilibrium trading of consumption across time?

Solution: The optimality condition with respect to bond holdings reads

$$0 = \frac{\partial U}{\partial b} = -(e_1 - b)^{-\gamma} + \beta E(e_2(s) + (1+r)b)^{-\gamma}(1+r)$$

The bond market clearing condition is given by

$$b = 0$$

as the bond is in zero supply. Thus, because all household's are identical in equilibrium there will be no trade of consumption across time, the interest rate will adjust such that agents consume exactly their income.

1.4 Part d

(5 Points) Consider the bond market clearing condition derived in part (c), what is the consumption of households in period one, c_1, and in the two states, $c_2(s_G)$ and $c_2(s_B)$, of period 2 then? Solution: Since the bond holdings are zero in equilibrium, the consumption in each state will simply be given by the corresponding income in the particular state

$$c_1 = e_1$$

$$c_2(s) = e_2(s), \forall s \in \{s_G, s_B\}$$

1.5 Part e

(10 Points) Use your results from part (c) to show that in equilibrium the gross interest rate of the bond is given by

$$1 + r = \frac{e_1^{-\gamma}}{\beta E[e_2(s)^{-\gamma}]} = \frac{e_1^{-\gamma}}{\beta[pe_2(s_G)^{-\gamma} + (1-\rho)e_2(s_B)^{-\gamma}]}$$

Solution: Rewrite the optimality condition in part c as (writing out the expectation operator)

$$\frac{(e1 - b)^{-\gamma}}{\beta[p(e_2(s_G) + (1+r)b)^{-\gamma} + (1-p)(e_2(s_B) + (1+r)b)^{-\gamma}]} = 1 + r$$

in equilibrium, where $b = 0$, this expression simplifies to

$$\frac{e_1^{-\gamma}}{\beta[pe_2(s_G)^{-\gamma} + (1-p)e_2(s_B)^{-\gamma}]} = \frac{e_1^{-\gamma}}{\beta E[e_2(s)^{-\gamma}]} = 1 + r$$

1.6 Part f

(10 Points) Assume now that $e_1 = 2$ in period 1, $e_2(s_G) = 3$ and $e_2(s_B) = 1$ in period 2, $\beta = 1/2$, $\gamma = 1$ and $p = 1/2$. What is the gross interest rate $1 + r$ in equilibrium? Solution: The equilibrium gross interest rate is given by

$$1 + r = \frac{1/2}{1/2[1/2 \times 1/3 + 1/2 \times 1]} = 3/2$$

1.7 Part g

(5 Points) Now assume instead that $e_1 = 2$ in period 1, $e_2(s_G) = 2$ and $e_2(s_B) = 2$ in period 2, $\beta = 1/2$, $\gamma = 1$ and $p = 1/2$. What is the gross interest rate $1 + r$ in equilibrium? Solution:

$$1 + r = \frac{1/2}{1/2[1/2 \times 1/2 + 1/2 \times 1/2]} = 4/2 > 3/2$$

1.8 Part h

(10 Points) Compare the equilibrium interest rate derived in part (g) to the one in part (f). Comment on your results and relate it to the precautionary savings motive that was discussed in class and the seminars. Solution: On average, the second period income under both scenarios is equal to the first period income. However, the scenario in part (f) features risk while there is no risk in part (g) concerning the second period income, and the interest rate is falling in the amount of risk. Since the marginal utility of the stated preferences is strictly convex, there is a precautionary savings motive for the households which is reflected by the fact that in the economy with risk (part (f)) the equilibrium interest rate is smaller than in the economy without risk, because the interest rate has to fall to accommodate the precautionary savings motive and to bring the bond market into the zero bond demand equilibrium.

2 University of Oslo Fall 2014 Exercise A.4

Consider the capital accumulation equation of the Ramsey model with exogenous technology growth

$$(1 + g)k_{t+1} = k_t^{\alpha} - c_t + (1 - \delta)k_t$$

Where $k_t := K_t/(A_t L)$, $c_t := C_t/(A_t L)$, and K_t is the aggregate capital stock, A_t is the state of technology, L the size of the population, C_t aggregate consumption, $\alpha \in (0, 1)$ the capital income share in the economy, $\delta \in (0, 1)$ the

depreciation rate of physical capital, and $g \geq 0$ denotes the net growth rate of technology. The Golden Rule capital stock per efficiency unit (the capital stock per efficiency unit that maximizes steady-state consumption per efficiency unit) is given by

$$k_{GR} = (\frac{\alpha\delta}{\delta + g})^{1/\alpha}$$

True or false?

Solution:

The Golden Rule capital stock is characterized by

$$k_{GR} = \max_{k \geq 0} k^{\alpha} - (\delta + g)k$$

with the associated optimality condition

$$0 = \alpha k_{GR}^{\alpha - 1} - \delta - g$$

$$k_{GR} = (\frac{\alpha}{\delta + g})^{1/(1-\alpha)}$$

Thus, the above statement is wrong.

3 University of Oslo Fall 2014 Exercise A.2

Consider the capital accumulation equation of the Solow model with exogenous technology growth

$$K_{t+1} = sK_t^{\alpha}(A_t L)^{1-\alpha} + (1 - \delta)K_t$$

where K_t is the aggregate capital stock, A_t is the state of technology, L the constant size of the population, s the exogenously given savings (investment) rate, $\alpha \in (0, 1)$ the capital income share in the economy, $\delta \in (0, 1)$ the depreciation rate of physical capital, $g \geq 0$ denotes the net growth rate of technology, and the capital stock per efficiency unit is defined as $k_t := K_t/(A_t L)$. The stable steady-state capital stock per efficiency unit, $k^* > 0$, of this Solow model is given by

$$k^* = (\frac{s}{\delta + g})^{1/(1-\alpha)}$$

True or false?

Solution:

First, detrend the capital accumulation equation by multplying by $1/(A_t L)$ to yield the accumulation equation in terms of efficiency units (remember $A_{t+1} = (1 + g)A_t$)

$$(1 + g)k_{t+1} = sk_t^{\alpha} + (1 - \delta)k_t$$

In the steady-state, the stable capital stock per efficiency unit will be constant, $k^* := k_{t+1} = k_t > 0$, such that the capital accumulation equation can be written as

$$(\delta + g)k^* = s(k^*)^\alpha$$
$$(k^*)^{1-\alpha} = \frac{s}{\delta + g}$$
$$k^* = (\frac{s}{\delta + g})^{1/(1-\alpha)}$$

Thus, the stated steady-state capital stock per efficiency unit is the correct one